Math 1200 Lecture 11

Oct 8 2026

Topics: 15.1-15.2: Double integrals, Fubini's theorem Name:_____
Collaborators:_____

Group work 1 Evaluate the following double integrals

a) Let R be the region defined by $0 \leq x \leq 3$, $-1 \leq y \leq 1$. Calculate $\iint_R x^2 + y^2 dA$.

b) Evaluate $\int_{-\pi}^{\pi} \int_0^3 e^{y^2} \sin(x) dy dx$

Group work 2 Set up, but do not evaluate, the integral of the function $f(x, y) = e^{x^2+y^2}$ on the region D which is the right half of the disk of radius 3 centered at the origin.

Group work 3 Let R be the quadrilateral with vertices $(0,0), (3,0), (2,1)$ and $(0,1)$. Evaluate $\iint_R f(x,y)dA$ for $f(x,y) = x + y$.

Group work 4 Evaluate $\int_0^1 \int_{\sqrt{x}}^1 \sqrt{y^3 + 4}\, dydx$

Group work 5 (Challenge problem) Find the volume between the paraboloid $z = 1 - x^2 - y^2$ and the xy-plane.